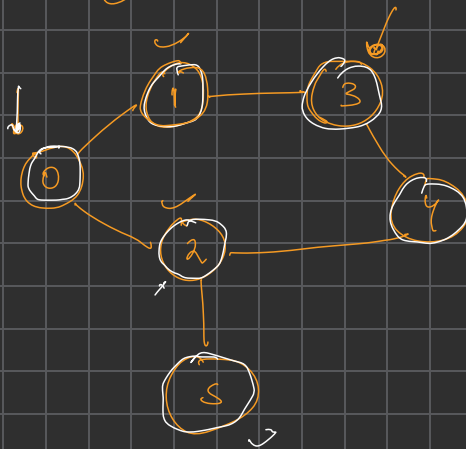


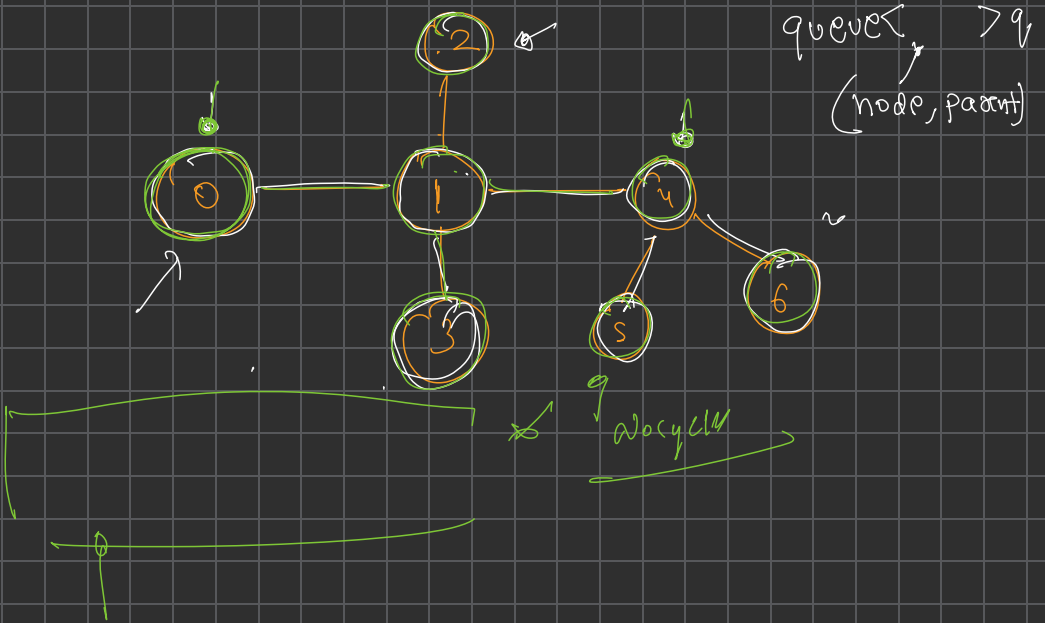


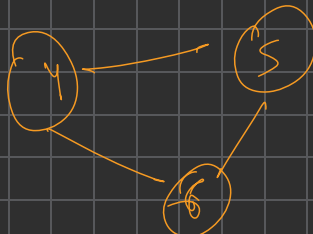
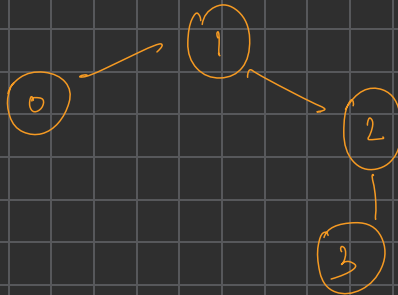
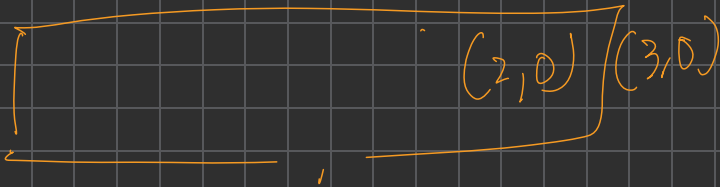
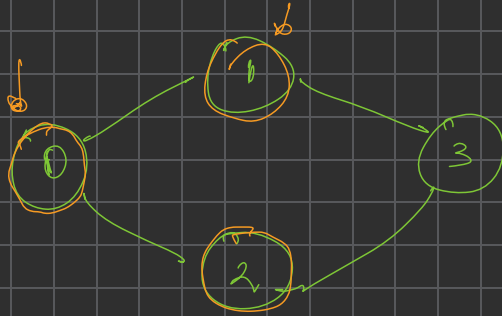
[BFS] $\Rightarrow$ 



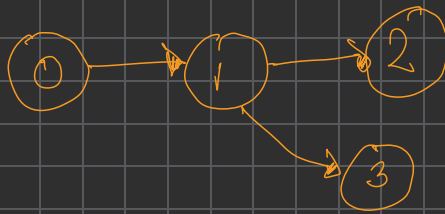
 Graph

Parent ko
ignore,
back mein
check
visited
aage cycle





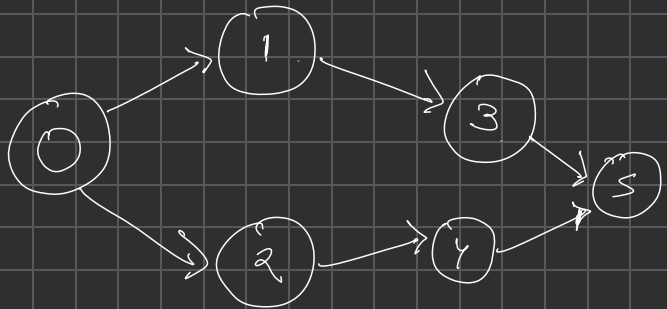
Topological Sort : DAG
 $U \rightarrow V$ Directed Acyclic Graph



Answer:

~~0 1 2 3~~

0 1 3 2



0 1 2 4 3 5

0 2 1 3 4 5

0 1 3 2 4 5 ✓

DFS

DAG

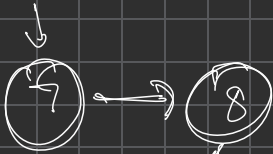
→ T.Sort



0 1 2 3 4

götte
hgyi

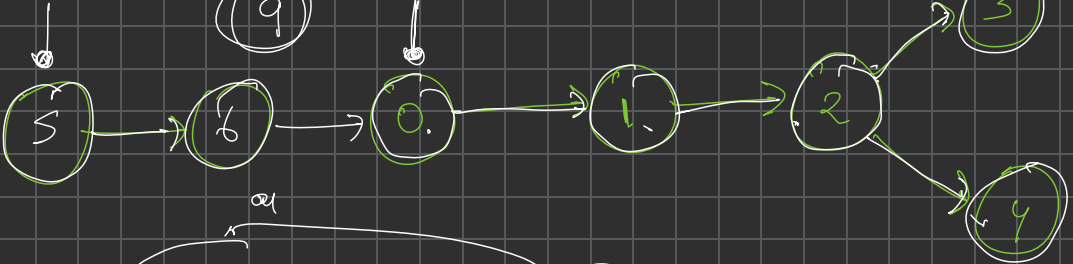
0	1	2	3	4	5	6
1	1	1	0	0	0	0



7 8 9



8 naen

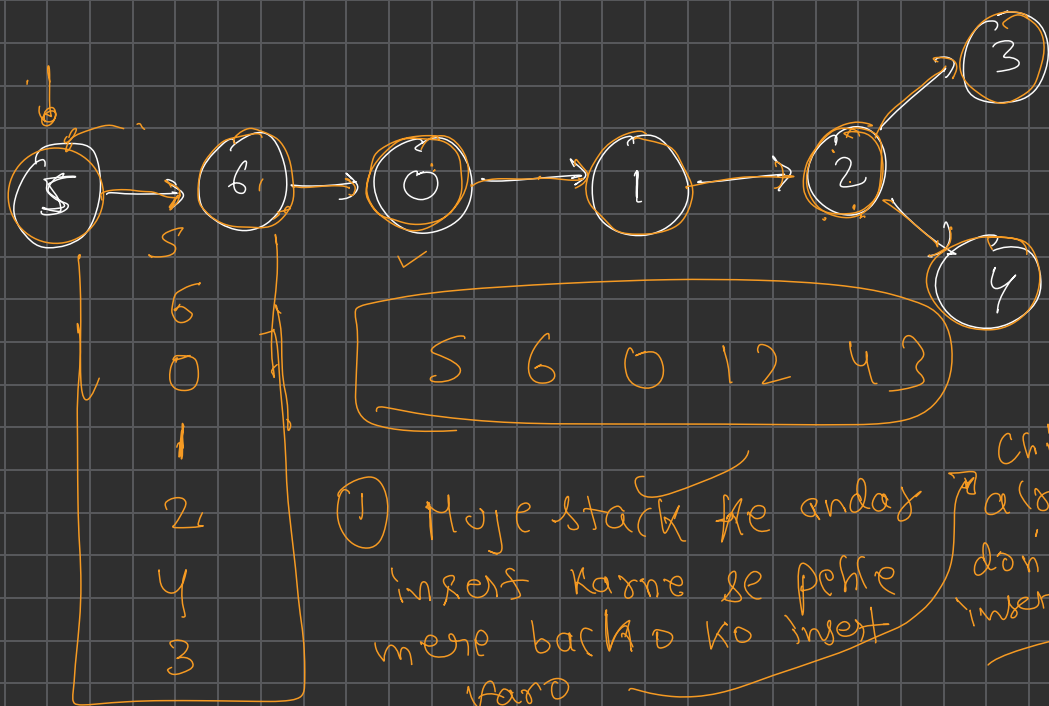


① ① 0 1 2 3 4 7 {0 1 2 3 4 5 6}

② ② 5 6 7 2nd [5 6 0 1 2 3 4]

③ 7 8 9 5 dr

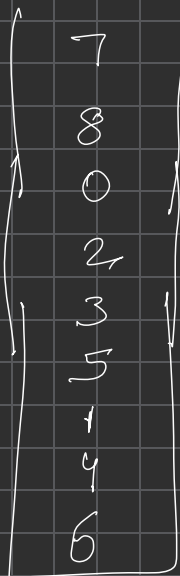
7 8 9 5 6 0 1 2 3 4



① Maje stack ke andar insert karne se pehle mere bacho ko insert karo

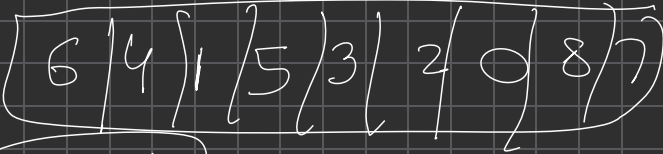
② the insert me

child already don't insert it

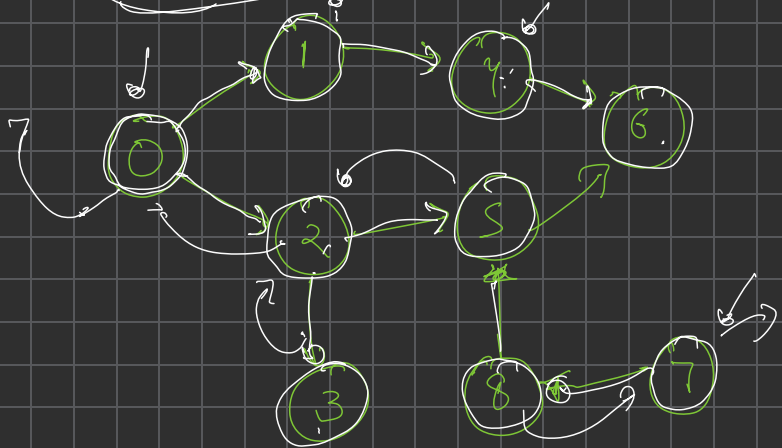


Stack

element
as chunk

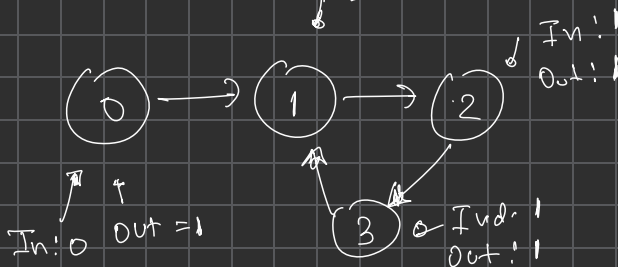


reverse



- ① Indegree : ? } Directed graph
② Outdegree :

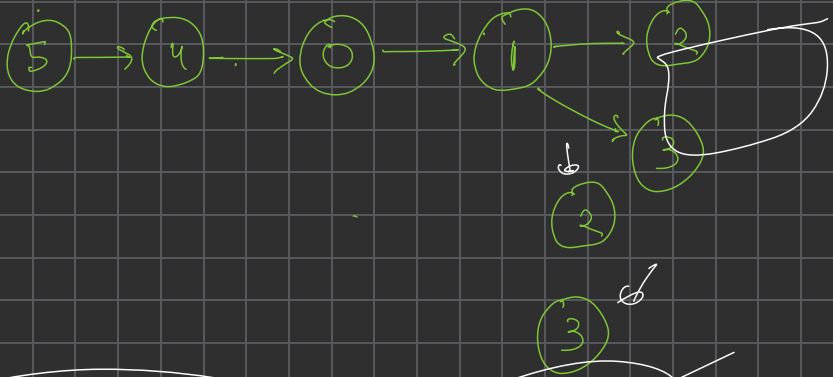
In: 2, Out: 1



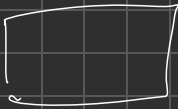
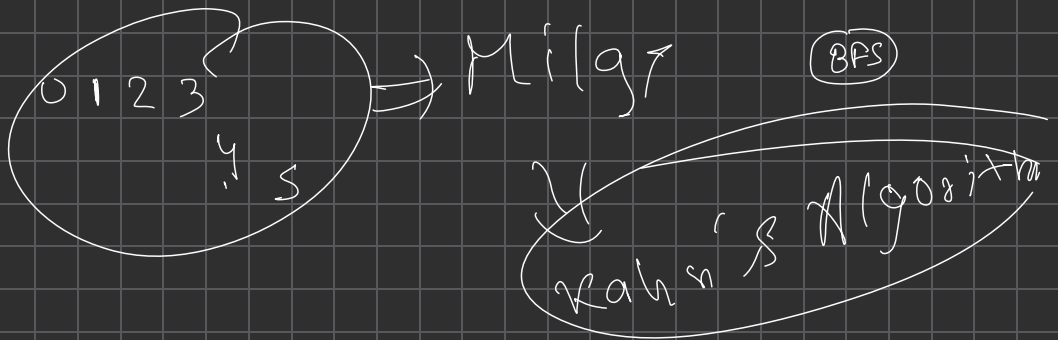
DA9

Kahn's Algorithm

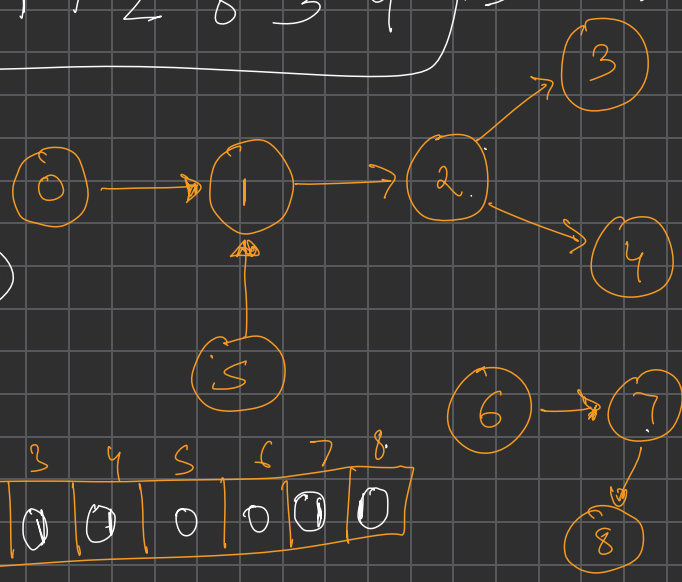
↳ Topological Sort



5 4 0 1 2 3 → output



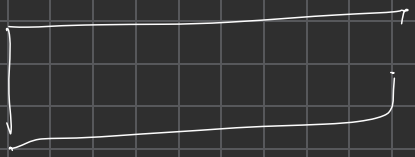
0 5 6 1 7 2 8 3 4 → answer K=1

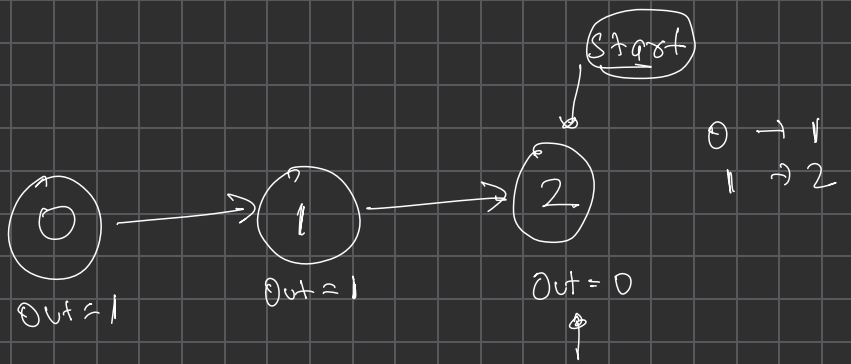


- ① Find index of each node
- ② Insert all the node into queue whose index < 200

Index

0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0



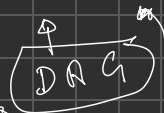


0 → 1
 1 → 2

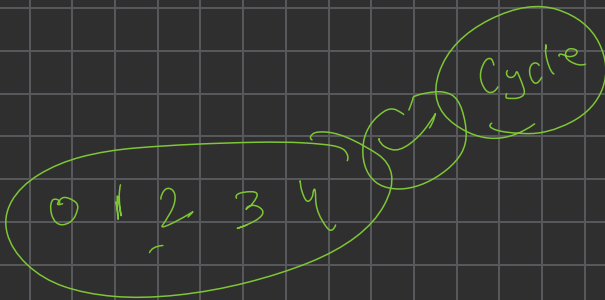
0	1	2
1	1	0

2

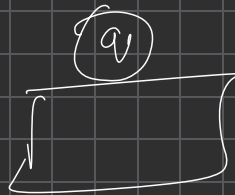
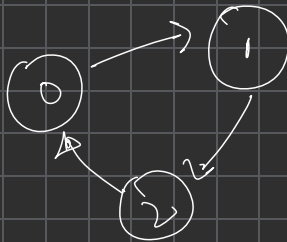
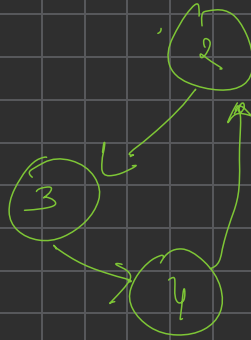
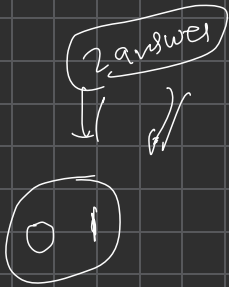
① Kahn's $\rightarrow$ Topological Sort



① Disjoint graph: Cycle



5



answer cycle present

result? →